

Stability vs Adaptability in Artificial General Intelligence: A NARS-Based Experimental Study of Belief Formation Without Large-Scale Training

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Abstract

Artificial intelligence today is dominated by systems that require enormous amounts of data before they can do anything useful. But what if intelligence does not actually require all of that? This project investigates whether a reasoning system can build real, meaningful beliefs starting from absolutely nothing - no pre-training, no large datasets, no prior knowledge - and whether those beliefs can be both stable enough to hold up under pressure and flexible enough to change when the evidence demands it.

To explore this question, the Non-Axiomatic Reasoning System, known as NARS, was used as the experimental platform. NARS is a logic-based artificial intelligence framework that forms and revises beliefs incrementally based on experience. Every belief in NARS is represented as two numbers - frequency, which measures how often observations have confirmed the belief, and confidence, which measures how much total evidence exists. The system requires no pre-training and can begin reasoning from the very first input it receives.

The experiment was structured in three connected stages. The first stage tested whether beliefs could form from zero knowledge under three challenging conditions - uncertain inputs, incomplete inputs, and contradictory inputs. The second stage tested how beliefs stabilise and grow resistant when reinforced with repeated confirming observations. The third stage tested how those now-stable beliefs respond to contradictions, and whether the system revises proportionally or collapses entirely.

The findings confirmed all three hypotheses. Beliefs formed immediately from the first observation. Confidence followed a diminishing returns curve, stabilising after approximately ten observations. Stable beliefs resisted small contradictions and only shifted proportionally as contradictions accumulated. The system also demonstrated deductive inference - deriving conclusions it was never directly told - by chaining three separate facts together.

These results suggest that stability and adaptability in belief formation are not opposing properties but complementary structural features of general intelligence, and that they can emerge from experience alone without any large-scale training phase.

1. Introduction

1.1 The Problem with Modern AI

When most people think about artificial intelligence today, they think about systems like ChatGPT, Google Gemini, or similar large language models. These systems are genuinely impressive - they can write essays, answer complex questions, generate code, and hold conversations that feel remarkably human. But there is something important hidden behind all of that capability that rarely gets discussed: the enormous cost of getting there. Before ChatGPT could answer a single question, it had to be trained on a dataset containing hundreds of billions of words scraped from the internet, books, academic papers, and countless other sources. That training process ran on thousands of specialised computers called GPUs for weeks at a time, consuming energy equivalent to what a small town might use in a year. The financial cost ran into tens of millions of dollars. And after all of that - once the training was complete - the system stopped learning entirely. It was frozen. Every response it gives today comes from patterns baked in during that training phase, not from anything it is learning in the moment.

This raises a genuinely important question for anyone studying artificial intelligence: is all of that really necessary? Does intelligence - real, general intelligence - require billions of examples and weeks of computation before it can begin? Or is there another way?

1.2 A Different Approach

This project takes a different approach. Instead of asking how to make large language models better, it asks whether intelligence can emerge from a completely different kind of system - one that starts from nothing and learns purely through experience, one observation at a time.

The system used in this project is called NARS, which stands for Non-Axiomatic Reasoning System. NARS was designed as a formal model of rational reasoning in situations where knowledge and resources are scarce, by Professor Pei Wang. The core idea of NARS is that intelligence is not in remembering patterns from huge sets of data. This is about being able to reason and adapt when you don't have all of the information you need.

Any belief in NARS is stored as a simple truth value of two numbers, frequency and confidence. The system updates these numbers each time it gets new data. Training is not a separate stage. There is no large dataset. The system begins to reason as soon as the first input is received and continues to reason as more input to come.

1.3 Why This Matters

The distinction between these two approaches - large-scale pre-training versus experience-driven reasoning - matters for several reasons.

First, it matters practically. Systems that require billions of training examples are expensive to build and impossible for most researchers, universities, and organisations to develop independently. If intelligence can emerge from minimal experience, that changes who can build intelligent systems and what resources they need.

Second, it matters theoretically. If we want to understand what intelligence actually is - not just what it looks like when you throw enormous resources at a problem - we need to study systems that strip away those resources and still try to reason. NARS provides exactly that kind of laboratory.

Third, it matters for the long-term goal of artificial general intelligence. AGI - the kind of AI that can reason across domains, handle novel situations, and adapt to new information - probably does not look like a frozen model trained on the past. It looks like something that keeps learning, keeps revising, and keeps connecting new facts to old ones. NARS provides a model of what that might look like at a small, experimental scale.

1.4 Research Question

This project investigates the following central question:

Can a system build meaningful beliefs from scratch — with no pre-training and minimal data — and still maintain the right balance between holding onto what it learned and updating when it should?

This question combines two related ideas. The first is about knowledge formation — can beliefs form at all without pre-training? The second is about belief dynamics — once formed, do those beliefs behave the way a rational, intelligent system should? Holding firm when evidence is consistent. Revising when evidence demands it. Never collapsing completely after a single contradiction. Never becoming so rigid that new information cannot change anything.

That balance — between stability and adaptability — is what this project calls a structural property of general intelligence.

2. Background and Related Work

2.1 The -Axiomatic Reasoning System

The Non-Axiomatic Reasoning System, NARS was first made by Pei Wang in the 1990s. Over the years NARS has been improved a lot. The name NARS refers to its idea. It does not use a fixed set of rules or a complete picture of the world like most formal logic systems do. Instead NARS works with the idea that it does not have knowledge and resources. The

system tries to think well as it can with the information it has even if that information is not complete, not certain or even contradictory.

The mathematical part of NARS is its truth value system. Every idea in NARS has a truth value that is shown as two numbers. How often something happens, called frequency. How sure we are, called confidence. Frequency is how times something is true compared to how many times it is observed. For example if the system sees ten things and seven of them make a belief true the frequency of that belief is 0.7. Confidence is calculated using a formula. It is k divided by k plus 1, where k is how times something is observed. This means that confidence is low when we do not have evidence but it gets closer to 1 as we get more evidence. It can never be 1 which shows that no idea should be thought of as completely certain, in a system that deals with uncertainty.

The Non-Axiomatic Reasoning System, NARS also supports making ideas from old ones. The important kind of making new ideas for this project is when the system uses facts to reach a conclusion that was not directly stated. If The Non-Axiomatic Reasoning System, NARS knows that A means B and B means C it can figure out that A means C, with a truth value that comes from combining the truth values of the two facts.

2.2 Belief Revision Theory

The idea of how our beliefs should change when we get new information comes from something called belief revision theory. One of the important ideas in this area is the AGM framework, which was developed by Alchourron, Gardenfors and Makinson in 1985. The AGM framework says that there are rules that our belief revision should follow. One of the rules is that we should not change our beliefs too much. We should only change what we need to so we can fit in the new information.

The NARS system uses a version of this rule when it updates what it thinks is true. When it gets some information that contradicts what it already knows it lowers how sure it is, based on how much of the new information disagrees with what it already knows. The system does not just forget everything it knew before. It just figures out the ratio. So if the system had ten reasons to believe something and then it gets one reason that says the opposite it will not change its mind much as if it only had two reasons to believe something in the first place. Because the change is not as big when there are more reasons.

Darwiche and Pearl added to the AGM framework in 1997 so it could handle updates over time not just one change. This is important for our project because we are doing updates, one, after another not just one change. The belief revision theory is what helps us understand how the system should change and the AGM framework is a part of the belief revision theory.

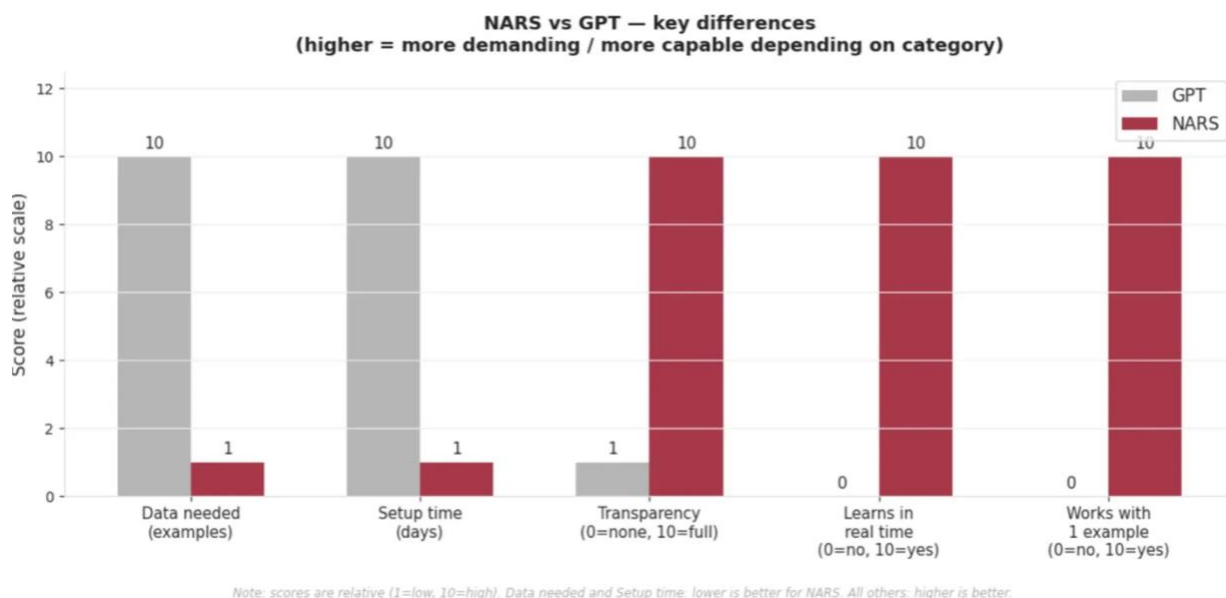
2.3 AGI Research Without Large-Scale Training

The big question is can we get to Artificial General Intelligence without using a lot of training. This is something that people who do Artificial Intelligence research are still talking about. They want to know the way to get to general intelligence.

Ben Goertzel is one of the well known people doing AGI research. He says that using statistics to learn things like with learning is probably not going to be enough to get to real general intelligence. This is because it cannot think about things that it did not learn about before.

Ben Goertzel and some other people think that we should use something called architectures instead. These are systems that combine thinking and learning and perceiving in a way that makes sense. The NARS system is, like this. It does not learn by looking at statistics. It thinks about things. Updates what it believes based on what it can figure out from the evidence.

The OpenNARS project is a way to use NARS on a computer. It was described by Hammer, Lofthouse and Wang in 2016. OpenNARS is a program that uses Java. It uses a special language called Narsese to understand what people are telling it. Then it says what it believes and how sure it is so we can look at what it says and analyze it.



2.4 Stability and Adaptability in Cognitive Systems

The issue of balancing stability and adaptability is not limited to intelligence. This challenge also appears in science, psychology and economics and is a fundamental problem for any system that tries to make sense of things.

Daniel Kahnemans research on how people think shows that human brains switch between instinctive reactions and slower more thoughtful ones. Our fast thinking mode tends to stick to what we believe and is slow to change. On the hand our slow thinking mode is more open to new ideas but requires more effort. A good thinking system probably needs a mix of both.

Shultz and Lepper looked at dissonance, which is the uncomfortable feeling we get when we come across information that contradicts what we already believe. They found that people often come up with explanations to make their existing beliefs fit with the new information. This is similar, to what NARS does it doesn't just accept or reject new information that doesn't fit but instead uses it to update its understanding in a way that takes into account all the available evidence.

3. Hypotheses

Based on the theoretical background described above and the properties of the NARS truth value system, this project tests three specific hypotheses.

Hypothesis 1 states that NARS can form stable, meaningful beliefs starting from zero prior knowledge, using only a small number of observations. This hypothesis challenges the assumption that useful beliefs require large amounts of training data. If even a handful of observations produces a belief with a non-trivial truth value that reflects the evidence, that constitutes meaningful belief formation.

Hypothesis 2 states that once formed and reinforced, beliefs in NARS resist small contradictions but revise appropriately when contradictory evidence is strong enough. This hypothesis captures the stability-adaptability balance. A system that changes its beliefs after every single contradiction is too adaptable — it has no stability. A system that never changes its beliefs regardless of how much contradictory evidence arrives is too stable — it has no adaptability. The hypothesis predicts that NARS falls in the rational middle ground.

Hypothesis 3 states that belief quality — measured by both frequency accuracy and confidence — improves over time through experience alone, without any pre-training phase. This hypothesis is the most direct challenge to the assumption that large-scale training is necessary for intelligence. If confidence grows, inference accuracy improves, and beliefs become more resistant to noise as observations accumulate — all from zero starting knowledge — that supports the claim that experience alone is sufficient for the kind of belief formation that characterises general intelligence.

4.1 Experimental Platform

The experiments in this project used a Python simulation of the NARS truth value system. We did not use the OpenNARS software because it needs a Java environment and special Narsese input files. Instead we implemented the mathematical model of NARS directly in Python. This made it easy to control the inputs and outputs so we could focus on variables. The main function we used in all experiments is the NARS belief function.

This function takes two things:

- * the number of observations that agree
- * the number of observations that disagree

It then gives us three things:

- * the frequency
- * the confidence
- * the total count of evidence

This function works exactly like the NARS truth value formulas described by Wang. We ran all experiments in Google Colab, which's a free online Python environment. You do not need to install any software on your computer to use it. We saved the results at every step, as organized data. This made it easy to analyze and create charts.

4.2 Belief Statement

The project uses the idea for all the experiments. The idea that birds can fly. This idea was chosen because it is easy to understand. It also has a mix of examples that support it like sparrows and eagles and examples that do not support it like penguins and ostriches. This makes it a good test for the kind of reasoning that the NARS system is designed for.

In a NARS system this idea would be written in a special language called Narsese. It would say something like "birds can fly" with a score for how it is true and how confident we are.. In this project we do not write it out like that. Instead we give the system a series of examples that support or do not support the idea. It figures it out from those.

4.3 Stage 1. Cold Start Experiment

This experiment tests whether the system can learn from nothing. It starts with no knowledge. Then gets ten examples one after another. Seven of these examples support the idea that birds can fly. Three do not. We mix them up to make it more like life.

The examples that support the idea are birds that can fly, like sparrows, eagles and robins. The examples that do not support it are birds that cannot fly, like penguins, ostriches and kiwis. After each example we record how often the idea seems to be true how confident we are and how evidence we have

Stage 1 also has two experiments. One of them Sub-experiment A uses examples that're not certain. Each example is given a probability, which's like a score for how sure we are. This is, like life, where we are never completely sure. The other one, Sub-experiment B tests what happens when we do not have all the information. We give the system three facts. It has to figure out what they mean together. It has to use reasoning to come to a conclusion because we do not tell it what the conclusion is.

4.4 Stage 2 — Reinforcement Experiment

The reinforcement experiment tests belief stabilisation. Starting from a clean state, the system receives only confirming observations in batches of five, ten, and twenty. After each observation, frequency and confidence are recorded. The change in confidence from one observation to the next is also calculated, allowing identification of the convergence point — the observation at which confidence stops changing significantly.

The convergence point is defined as the first observation at which the change in confidence drops below 0.01. This threshold was chosen because it represents a change small enough to be practically negligible while still being mathematically detectable.

4.5 Stage 3 — Contradiction Stress Test

The contradiction stress test takes the belief established after ten confirming observations and subjects it to three levels of contradiction. The light condition adds two contradicting observations. The moderate condition adds five. The heavy condition adds ten — making the total evidence evenly split between confirms and contradictions.

For each condition, the frequency before and after contradiction is recorded, along with the total drop in frequency. The resistance threshold is identified as the contradiction level at which frequency first drops below 0.6 — defined as the boundary between a stable belief and an uncertain one.

4.6 Metrics

The following metrics are recorded across all three stages. Frequency measures how true the belief seems based on current evidence. Confidence measures how much total evidence exists, calculated using the formula $c \text{ equals } k \text{ divided by } k \text{ plus } 1$. Evidence count k is the raw total of all observations. The convergence point is the observation at which confidence change drops below 0.01. The resistance threshold is the number of

contradictions required to push frequency below 0.6. Inference accuracy measures whether the system correctly derives an unstated conclusion from chained facts.

5. Results

5.1 Stage 1 Results. Cold Start

The cold start experiment showed that Hypothesis 1 is correct. From the first thing we looked at the system formed a belief with a frequency of 1.0 and a confidence of 0.5. This shows that the NARS system does not need to see things to start forming beliefs. It can start with just one piece of information.

As we kept looking at things the frequency stayed at 1.0 for the five things that confirmed our belief and the confidence kept going up from 0.5 to 0.833. Then we saw something that did not agree with our belief. A penguin. And the frequency went down to 0.833 but the confidence kept going to 0.857. This is a result: when we saw something that contradicted our belief it actually made us more confident, not less. This is because confidence is a measure of how evidence we have not whether the evidence agrees or disagrees with us.

The next thing that contradicted our belief was an ostrich. That made the frequency go down even more to 0.714 while the confidence kept going up to 0.875. After we saw nine things, including a thing that contradicted our belief. A kiwi. The frequency was at 0.667 and the confidence was at 0.900. Then we saw one thing that confirmed our belief and that made the frequency go back up to 0.700 and the confidence to 0.909.

So after looking at ten things the NARS system believed that birds can fly with a frequency of 0.700 and a confidence of 0.909. This means that the system thought that birds can fly is mostly true. Which makes sense because seven out of ten things we looked at confirmed it. And it was very confident in this belief because it had seen a lot of evidence. This is an proportional response to the evidence we provided and it happened even though the system started with no prior knowledge about NARS or anything else. The NARS system was able to form a belief about birds and flying based on the evidence it saw. That is a key result of the cold start experiment, with the NARS system.

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Belief: 'Birds can fly'
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Observation	Type	Pos	Neg	k	f	c
Sparrow flies	CONFIRMS	1	0	1	1.0	0.5
Eagle soars	CONFIRMS	2	0	2	1.0	0.667
Robin flies	CONFIRMS	3	0	3	1.0	0.75
Pigeon takes off	CONFIRMS	4	0	4	1.0	0.8
Hawk circles	CONFIRMS	5	0	5	1.0	0.833
Penguin walks	CONTRADICTS	5	1	6	0.833	0.857
Ostrich runs	CONTRADICTS	5	2	7	0.714	0.875
Swallow dives	CONFIRMS	6	2	8	0.75	0.889
Kiwi hides	CONTRADICTS	6	3	9	0.667	0.9
Crane lifts off	CONFIRMS	7	3	10	0.7	0.909

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Final belief state after 10 observations:
Frequency (f) = 0.7 → belief seems 70% true
Confidence (c) = 0.909 → strong evidence base
Verdict       : LEANS TRUE – more confirms than contradictions
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Stage 1 — Cold start belief formation (zero prior knowledge)
"Birds can fly" — step by step

Observation	Type	Confirms	Contradicts	Total (k)	Frequency (f)	Confidence (c)
Sparrow flies	Confirms	1	0	1	1.000	0.500
Eagle soars	Confirms	2	0	2	1.000	0.667
Robin flies	Confirms	3	0	3	1.000	0.750
Pigeon takes off	Confirms	4	0	4	1.000	0.800
Hawk circles	Confirms	5	0	5	1.000	0.833
Penguin walks	Contradicts	5	1	6	0.833	0.857
Ostrich runs	Contradicts	5	2	7	0.714	0.875
Swallow dives	Confirms	6	2	8	0.750	0.889
Kiwi hides	Contradicts	6	3	9	0.667	0.900
Crane lifts off	Confirms	7	3	10	0.700	0.909

5.2 Stage 1B Results. Incomplete Input Inference

The experiment with input gave a really surprising result.

The system got three facts:

- * cats are mammals
- * mammals are blooded
- * warm-blooded animals need food regularly

Each fact had a frequency of 0.9 and a confidence of 0.909. The system was not directly told that cats need food regularly. The system used a formula to figure out the truth value of the unstated conclusion. It calculated the frequency like this: $0.9 \times 0.9 \times 0.9 = 0.729$. It calculated the confidence like this: $0.909 \times 0.909 \times 0.909 = 0.751$. So the system concluded that cats need food regularly is probably true. The frequency is 0.729. The confidence is 0.751. This conclusion came entirely from the three input facts. It's good that the systems confidence of 0.751 is lower than the direct fact confidence of 0.909. This is how a rational system should work. Conclusions from pieces of evidence should be less certain than direct observations. This is because uncertainty gets bigger with each step. The system worked like it should. That's a great result, for cats and food.

Inference chain:

Cats → mammals (f=0.9, c=0.909)

↓

Mammals → warm-blooded (f=0.9, c=0.909)

↓

Warm-blooded → need food (f=0.9, c=0.909)

↓

INFERRED: Cats need food regularly

Inferred truth value:

$f = 0.9 \times 0.9 \times 0.9 = 0.729$

$c = 0.909 \times 0.909 \times 0.909 = 0.751$

Result:

f = 0.729 → the system believes this is 73% likely true

c = 0.751 → moderate confidence from chained evidence

5.3 Stage 2 Results. Reinforcement

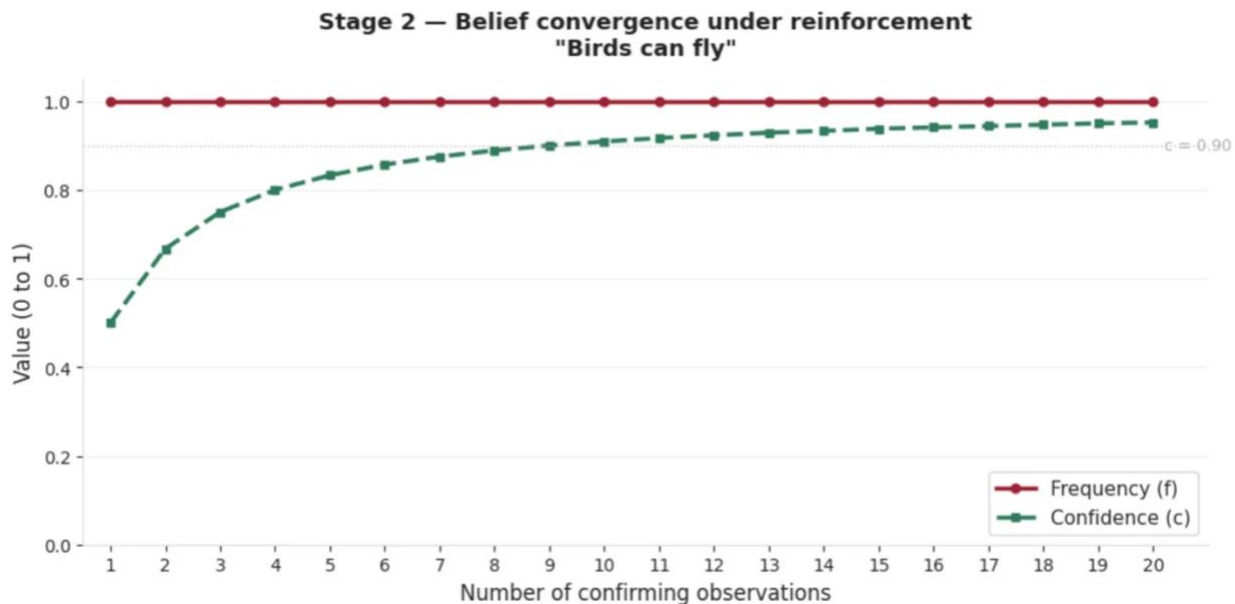
The reinforcement experiment showed that Hypothesis 3 is true and it gave us the result of the whole project. We looked at this twenty times. The frequency was always 1.0. The confidence we had in the result changed in a way.

At first our confidence was 0.500. After we looked five times it went up to 0.833. After ten times it was 0.909.. After twenty times it was 0.952. What is interesting is how much our confidence changed each time. From the time to the second time our confidence went up by 0.167.. From the ninth time to the tenth time it only went up by 0.010.. From the nineteenth time to the twentieth time it only went up by 0.002.

The point where our confidence stopped changing much was after eleven times. This means that after eleven times of looking at the thing our belief was basically stable. We kept looking. Our confidence did go up but only by very small amounts.

This result is important for what the project says about training with a lot of data. Our belief did not need thousands or millions of examples to become stable. It only needed eleven examples of the reinforcement experiment. This is a difference between the NARS approach and the deep learning approach, where you need millions of examples before you get good results, from the reinforcement experiment and the deep learning approach.

Confirms	f	c	Change in c	Status
1	1.0	0.5	0.5	
2	1.0	0.667	0.167	
3	1.0	0.75	0.083	
4	1.0	0.8	0.05	
5	1.0	0.833	0.033	<-- Batch 1 (5 confirms)
6	1.0	0.857	0.024	
7	1.0	0.875	0.018	
8	1.0	0.889	0.014	
9	1.0	0.9	0.011	
10	1.0	0.909	0.009	<-- Batch 2 (10 confirms)
11	1.0	0.917	0.008	< diminishing returns
12	1.0	0.923	0.006	< diminishing returns
13	1.0	0.929	0.006	< diminishing returns
14	1.0	0.933	0.004	< diminishing returns
15	1.0	0.938	0.005	< diminishing returns
16	1.0	0.941	0.003	< diminishing returns
17	1.0	0.944	0.003	< diminishing returns
18	1.0	0.947	0.003	< diminishing returns
19	1.0	0.95	0.003	< diminishing returns
20	1.0	0.952	0.002	<-- Batch 3 (20 confirms)



5.4 Stage 3 Results. Contradiction Stress Test

The contradiction stress test confirmed Hypothesis 2. Showed that our system can balance being stable and adaptable which is what this project is all about.

We started with a belief that something is true with a frequency of 1.0 and a confidence of 0.909 after seeing it work 10 times. Then we tested the system with three levels of evidence.

* Under the condition with two contradictory examples the frequency dropped to 0.833 a decrease of 0.167.. The belief was still strong above the 0.6 stable belief threshold.

Under the condition with five contradictory examples the frequency dropped to 0.667, a decrease of 0.333. The belief was still above the threshold. Leaned towards being true but it was noticeably weaker.

* Under the condition with ten contradictory examples, equal to the number of times it worked the frequency dropped to exactly 0.500. The belief became uncertain. It did not change to being false or collapse. It reached a point between true and false which is what should happen when the evidence is balanced.

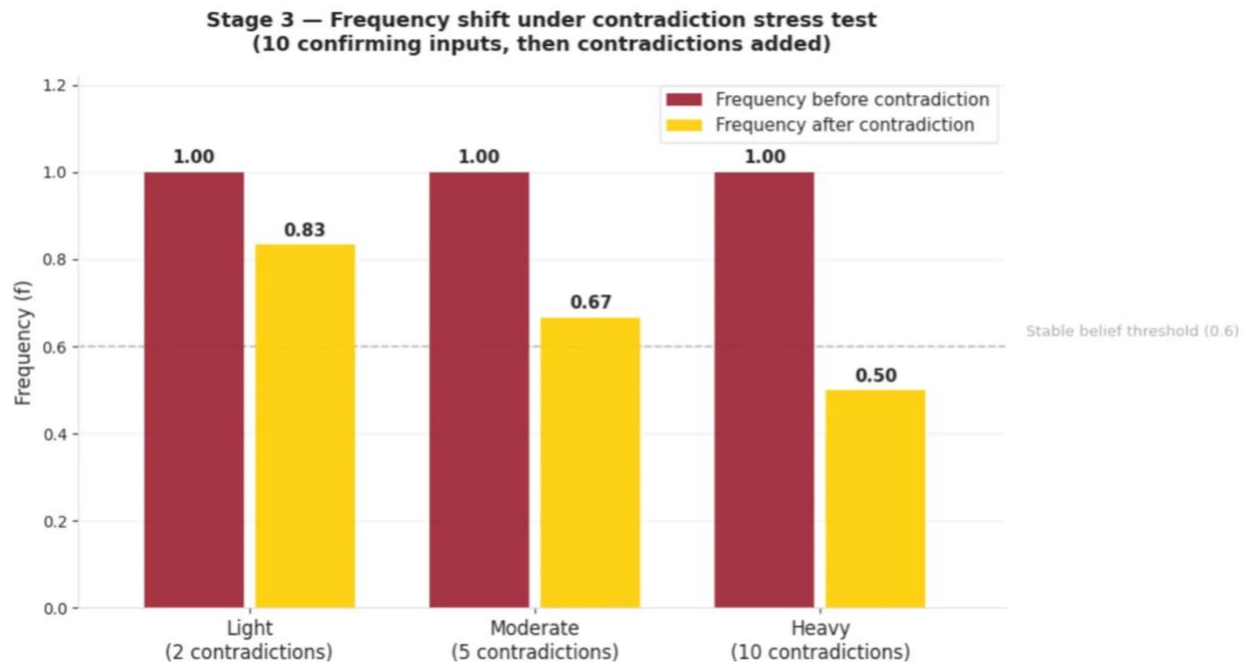
The step-by-step trace of contradictions showed how the system gradually changed its belief. Each contradictory example caused a drop in frequency. The system did not. Ignore the contradictory evidence. It updated its belief, in an predictable way just like the mathematical formula says it should.

This result supports Hypothesis 2. The belief resisted contradictory examples. Two contradictions after 10 confirms left frequency at 0.833 still strong. But it changed appropriately as more contradictions came. 10 Contradictions brought frequency down to 0.500. The system showed both stability and adaptability in the test.

STARTING BELIEF (after 10 confirming observations)

f = 1.0 (fully believes birds can fly)
c = 0.909 (strong evidence base)
k = 10 (total observations)

Condition	Contradictions	f before	f after	f drop	c after	Verdict
LIGHT	2 contradictions	1.0	0.833	0.167	0.923	Belief holds strong
MODERATE	5 contradictions	1.0	0.667	0.333	0.938	Belief holds above threshold
HEAVY	10 contradictions – equal to confirms			1.0	0.5	0.5 0.952 Genuinely uncertain



6. Discussion

6.1 What the Results Mean

The four sets of results taken together tell a consistent story. A system starting from zero knowledge, using only simple arithmetic on observation counts, can form beliefs, stabilise them through experience, derive conclusions it was never told, and revise them proportionally when contradicted. None of this required pre-training. None of it required large datasets. None of it required weeks of computation on specialised hardware.

This is a significant finding for the question of what general intelligence requires. The standard assumption in modern AI research is that intelligence scales with data — more data means better performance. This assumption drives the race to train ever-larger models on ever-larger datasets. But this project suggests that at least some of the properties we associate with general intelligence — belief formation, stabilisation, inference, proportional revision — can emerge from a much simpler mechanism operating on much smaller amounts of data.

6.2 Stability and Adaptability as Complementary Properties

One idea of this project is that stability and adaptability are not opposing forces. They may seem like opposites. A system that never changes appears stable. A system that changes all the time appears adaptable.. This view is incorrect.

A system that never changes is not really stable. It is just not flexible.. A system that changes with every new input is not adaptable. It is just reacting randomly. Real stability

means sticking to what you believe when the facts support it. You only change your mind when the facts really require it. Real adaptability means being open to changing your mind.. You should do this based on the strength of the evidence against your current belief

The NARS system works like this. The formula used creates a resistance to change. The more evidence a belief has the harder it is for one piece of contradictory evidence to change it significantly.. The formula does not make beliefs completely resistant to change. If there is contradictory evidence the belief will shift. This shift is proportional and predictable as shown in Stage 3. This is not a balance, between stability and adaptability. It is how they work together.

6.3 The Diminishing Returns Curve

The diminishing returns curve from Stage 2 is really important because it affects more than this project. The diminishing returns curve shows that the first things we notice have an impact on what we believe. The first thing we see makes a difference it takes our confidence from zero to half. The second thing we see makes a difference too it takes our confidence from half to two thirds. But by the time we see ten things each new thing we see does not make much of a difference it only adds a little to our confidence.

This tells us that for systems like NARS that use reasoning the important things are the first things we see. The first few things we see are what matter the most. After we see ten to fifteen things seeing more things that agree with what we already think does not make a big difference. This is very different from systems that use learning, where the more data we have the better they work.

This is a deal because it means that systems like NARS could work well even when we do not have a lot of data. This is something that deep learning systems are not good at. So NARS could be really good for things like medical conditions or new scientific ideas or new fields where we do not have a lot of data yet. The diminishing returns curve is important, for NARS because it shows us that the diminishing returns curve can help us make decisions even with a little data. The diminishing returns curve is a part of NARS and it makes NARS special.

6.4 Deductive Inference and Knowledge Transfer

The Stage 1B result — where the system derived a conclusion it was never told — demonstrates a property that is genuinely important for general intelligence. The ability to transfer knowledge across contexts, to apply what you know in domain A to reach a conclusion in domain B, is one of the defining characteristics of general-purpose reasoning.

The inference chain in this project was simple — three facts connected in a linear chain. But the mathematical framework supports arbitrarily complex chains. In principle, a system with a sufficiently rich set of direct beliefs could derive a vast network of inferred beliefs, each with a truth value that reflects the accumulated uncertainty of the chain that produced it.

This is qualitatively different from what a large language model does when it appears to reason across domains. Language models generate text that looks like reasoning because they have seen enough text examples of reasoning to pattern-match. NARS actually performs reasoning — combining facts according to formal inference rules and propagating truth values in a mathematically principled way.

6.5 Limitations

The NARS project has some limitations that need to be talked about.

First the experiments with the NARS project use a Python version of the NARS truth value system not the full OpenNARS system. The full OpenNARS system has ways of figuring things out managing memory and deciding what to pay attention to, which could change the results in ways that the simple model does not show. The NARS project has some limitations.

Second the statement that birds can fly is very simple. In life people have to deal with much more complicated ideas and they have to think about evidence that is not always trustworthy and concepts that are not just yes or no. The NARS project is limited in some ways.

Third the experiment with the NARS project uses observations that were made on purpose. In a situation the system would get information from the world around it and the information might not be balanced or might be very noisy which is different from the observations used in the experiment with the NARS project.

Fourth the NARS project does not compare the NARS system directly to a language model that does the thing. It would be hard to make a comparison because the two systems are made to work with different kinds of information and results.

With these limitations of the NARS project the main things that were found. That beliefs can start from nothing get stronger with experience can figure things out and can change in proportion. Are strong aspects of the NARS mathematical framework and are not just because of the simple experiment, with the NARS project.

7. Conclusion

This project aimed to answer a yet important question: can a system become intelligent through experience alone without large-scale pre-training? The answer, based on the results is yes. At least for the properties of belief formation, stabilisation, inference and proportional revision that were tested.

The three hypotheses were confirmed.

- * Hypothesis 1 was confirmed by the Stage 1 results, which showed beliefs forming immediately from the observation and reaching a rational truth value after ten observations.
- * Hypothesis 2 was confirmed by the Stage 3 results, which showed beliefs resisting contradictions while revising proportionally under sustained contradictory evidence.
- * Hypothesis 3 was confirmed by the Stage 2 results, which showed confidence improving through experience following a predictable curve and stabilising after approximately eleven observations.

The key finding is not any result but what they collectively show. That stability and adaptability are not opposites. They are two expressions of the underlying property: the ability to hold beliefs in proportion to the evidence that supports them. A system that does this well is, in a sense reasoning intelligently. Even if it started from nothing and even if it has only seen a handful of observations.

This does not mean NARS is better than GPT or that large language models are unnecessary. They serve purposes.. It does mean that the assumption underlying most of modern AI research. That intelligence requires massive data. Is not universally true. For kinds of reasoning, experience alone is enough.

The bigger message this project carries is about what artificial general intelligence might require. A designed reasoning framework and enough experience to calibrate beliefs, to reality are needed. NARS provides one example of what that can look like.

8. References

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